Draft v12 Revision Scope

Revision made

The revision focused on substantive report corrections requested after the latest supervision meeting: strengthening the Literature Review, adding missing implementation detail, correcting wording that was stronger than the available evidence, separating offline and online experimental settings, and improving repository traceability. Language-polishing work is not counted here as a technical revision.

Rationale

This structure makes the revision auditable. If the supervisor or examiner compares versions, Draft v12 can be explained as a targeted correction pass rather than an uncontrolled rewrite. Keeping the focus on evidence, reproducibility, and scope also reduces the risk that the final report overclaims beyond what the project files and experiment outputs can support.

Literature Review: Context and Dataset Choice

Revision made

The Literature Review was expanded to explain the system structure before the architecture diagram appears in the Methods chapter. A new discussion links preprocessing, MI decoding, and RL-based control as connected parts of the same BCI-to-control pipeline. The dataset section was also expanded to explain why BCI IV-2a, BCI IV-2b, and PhysioNet EEGMMIDB were selected instead of adding further public EEG datasets.

Rationale

This change addresses two likely reader questions before they become problems: what the architecture diagram represents, and why these datasets are sufficient for the project aims. BCI IV-2a gives a standard 4-class benchmark, BCI IV-2b tests minimal-channel decoding, and PhysioNet provides the scale needed for cross-subject pre-training. The review now supports the methodological choices rather than simply listing background work.

Text Comparison: Literature and System Context

Draft v11

The system is organised into three visual groups that together realise four integrated functions: EEG preprocessing, deep learning classification, reinforcement learning control, and hardware integration.

Draft v12

The system is organised into three visual groups that correspond to four functions: EEG preprocessing, deep learning classification, reinforcement learning control, and hardware integration.

Draft v11

The figure was introduced directly in Methods, so the reader had to infer how the literature review connected to the architecture diagram.

Draft v12

The new Literature Review bridge explains that the architecture diagram connects preprocessing literature, hybrid CNN–Transformer decoding, and RL-based decision making into one BCI-to-control pipeline.

Why this is clearer

The comparison shows that v12 did not just add background text; it changed the report flow so the architecture diagram is motivated before it appears.

Methods: Reproducibility and Code-Level Detail

Revision made

The Methods chapter now gives concrete implementation parameters for the EEGTransformer and the DQN control models. The report specifies input shapes, convolution settings, embedding size, token count, Transformer depth and heads, classifier head dimensions, optimiser settings, augmentation settings, and DQN replay/training parameters. It also adds code-level notes identifying the main project scripts for CTNet modelling, PhysioNet training, fine-tuning, channel reduction, and RL control.

Rationale

The previous draft described the models at a higher level, which made the work harder to reproduce and easier to question. Listing the parameters makes clear that the report corresponds to the implemented code rather than to a generic CNN–Transformer description. The code-level notes also help the examiner connect the written methodology to the repository without needing to infer which scripts produced which results.

Text Comparison: Methods Reproducibility

Draft v11

The architecture table listed the main layers, but did not explicitly state input shape, spatial feature maps after depthwise convolution, positional encoding, feed-forward expansion, classifier head, or residual fusion.

Draft v12

Added rows include: input shape $1 \times 64 \times 1000$, $F_2 = 40$, learnable positional encoding, $4\times$ feed-forward expansion, Dropout 0.5 $+$ Linear $600 \rightarrow 4$, and residual fusion.

Draft v11

The main PhysioNet script pools all available training participants and applies a stratified 80/20 split at the epoch level.

Draft v12

For the reported 109-participant PhysioNet experiment, the pooling script was run with an explicit full-subject list rather than its default `--subjects 1 2 ... 10` setting.

Why this is clearer

The revised text makes the method reproducible from the report itself, rather than requiring the examiner to reconstruct key parameters from code or figures.

Experimental Consistency: Time Windows and Training Setup

Revision made

Draft v12 separates the offline PhysioNet epoch definition from the intended online control window. The offline PhysioNet loader is now described as extracting epochs from -0.5 s to 4.0 s around cue onset, giving 4.5 s total before resampling to 1000 samples. The real-time control design is separately described as using a 4.0 s sliding decision window. The PhysioNet pooled pre-training description was also corrected to state that the reported 109-subject run used an explicit full-subject list rather than relying on the script default.

Rationale

This avoids mixing two different experimental settings. The 4.5 s offline window belongs to model development on PhysioNet, while the 4.0 s window belongs to the online-control design. Separating them prevents a reviewer from thinking that the report contradicts the loader code. Clarifying the explicit 109-subject run also distinguishes final experimental configuration from script defaults, which is important for reproducibility.

Text Comparison: Offline and Online Windows

Draft v11

In this context, "long-range temporal dependencies" refers to relationships spanning different parts of the same 4-second imagery trial.

Draft v12

In this context, "long-range temporal dependencies" refers to relationships spanning different parts of the same offline PhysioNet epoch, extracted from -0.5 s to 4.0 s relative to cue onset (4.5 s total).

Draft v11

BrainFlow streams EEG from the OpenBCI board, a 4-second sliding window is filtered and normalised online, the personalised EEGTransformer predicts class and confidence.

Draft v12

This 4.0 s online decision window is a deployment-oriented design choice and should be distinguished from the 4.5 s offline PhysioNet epochs.

Why this is clearer

The before/after wording makes the correction visible: v12 prevents the offline model-development window and online decision window from being read as the same setting.

Channel Reduction and OpenBCI Deployment Claims

Revision made

The channel-reduction section now lists the exact retained electrodes. The ablation-ranked 8-channel subset is reported as {C3, CP3, Fpz, Pz, O1, F8, FCz, FC2}, while the domain-guided OpenBCI-oriented subset is reported as {C3, C4, FC3, FC4, CP3, CP4, Cz, FCz}. The discussion was also narrowed so that the reported 8-channel result is presented as offline evidence, while the current online BrainFlow path is described as adapting low-channel streams to the legacy CTNet input shape.

Rationale

Naming the channels is necessary because "reduced to 8 channels" is not enough for replication or hardware planning. The wording is also more defensible: the offline 8-channel model supports the feasibility of an OpenBCI-compatible montage, but it does not prove a native end-to-end 8-channel online deployment. This distinction prevents the report from overstating the maturity of the hardware path.

Text Comparison: 8-Channel Claim Scope

Draft v11

The 8-channel motor cortex configuration (C3, C4, FC3, FC4, CP3, CP4, Cz, FCz), designed for OpenBCI Cyton compatibility, achieved 40.7% in cross-subject mode.

Draft v12

The 8-channel motor cortex configuration (C3, C4, FC3, FC4, CP3, CP4, Cz, FCz), designed as an OpenBCI Cyton-compatible electrode set for offline evaluation, achieved 40.7% in cross-subject mode.

Draft v11

These results suggest that the 8-channel configuration remains a plausible option for closed-loop control, although that claim still needs to be verified with real-time OpenBCI recordings.

Draft v12

These results suggest that the 8-channel electrode set remains a plausible offline foundation for closed-loop control, although real-time OpenBCI validation and a native low-channel online inference path remain future work.

Why this is clearer

The revised phrasing keeps the useful 8-channel evidence, but removes the implication that a complete native 8-channel online deployment has already been validated.

Hardware Validation and Architecture Diagram

Revision made

Hardware validation wording was revised to state that the BrainFlow acquisition interface and online control loop were exercised with synthetic-board streaming and archived EEG-driven simulation runs. Claims of verified live OpenBCI EEG streaming were removed. The architecture diagram source was also updated so the preprocessing block refers to fixed-length epochs rather than a universal 4 s epoch, keeping the figure consistent with the distinction between offline and online windows.

Rationale

The available evidence supports software-interface validation and simulation-level integration, not live testing with a real OpenBCI headset and new human subjects. The revised wording therefore matches the saved evidence files and the ethical-approval limitation. Updating the standalone architecture diagram avoids a second source of inconsistency, since figures are often inspected more quickly than the surrounding text.

Text Comparison: Hardware Validation

Draft v11

The BrainFlow API integration was implemented and tested for real-time EEG streaming from OpenBCI boards.

Draft v12

The BrainFlow acquisition interface and online control loop were exercised with synthetic-board streaming and archived EEG-driven simulation runs.

Draft v11

All evaluation in this project was conducted in simulation using pre-recorded EEG datasets.

Draft v12

EEG classification, preprocessing, and channel-reduction experiments were evaluated offline using pre-recorded public datasets. The end-to-end control evaluation therefore simulates real-time use.

Why this is clearer

The revised text now matches the available evidence: software-interface and simulation validation are supported, while live OpenBCI human-subject validation is left as future work.

Experiment Evidence and Repository Traceability

Revision made

Missing experiment outputs from the older CTNet workspace were migrated into `03_Experiments/`. This includes the authoritative architecture-comparison results, dataset-comparison logs, BrainFlow physical-control outputs, and real-time EEG-control outputs. A new `03_Experiments/README.md` was added to identify which files support the report and to reduce confusion caused by older or alternative result folders.

Rationale

This change improves traceability. The report now points to result files that exist inside the submitted repository rather than relying on paths left in an older working directory. The README is also useful because several versions of experiment outputs remain in the repository; without an index, a reviewer could open an older summary file and think the report numbers are unsupported or inconsistent.

Project Management and GitHub Documentation

Revision made

The Project Management section was expanded to describe planning, scope control, risk management, supervision checkpoints, and the reason for using simulation-first validation. The GitHub README and supporting repository descriptions were also updated so that the main project structure, report location, experiment outputs, and documentation folders have English explanations suitable for external inspection.

Rationale

This addresses assessment criteria beyond model accuracy. The expanded management section shows how technical risk was handled during the project, especially the decision to avoid unsupported live-subject claims and to keep hardware work as software-interface validation. The README updates make the repository easier for the supervisor or examiner to navigate, which supports the report's claim that the implementation is reproducible and inspectable.